

Technology Stack

Date	10/07/2026
Name	Jaswanth Madapuram
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, Jinja2 Templates	Provides a responsive and user-friendly interface for entering applicant details and displaying prediction results. Jinja2 enables dynamic rendering of prediction output.
2	Backend / Server-Side	Python 3, Flask	Flask handles HTTP requests, processes user input, communicates with the ML model, and returns prediction results. Python provides strong support for machine learning.
3	Database / Data Storage	CSV Dataset (credit_card_data.csv)	Stores historical applicant records used for model training. Lightweight and suitable for machine learning experiments and academic projects.
4	Cloud / Hosting / Deployment	Localhost (Flask Development Server) (Can also be deployed on Render or Heroku)	Used for testing and demonstrating the application. Easy deployment and maintenance with minimal configuration.

5	Version Control & CI/CD	Git & GitHub	Enables version control, collaborative development, code backup, and project management.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Joblib, Matplotlib, Seaborn	Scikit-learn builds and evaluates the prediction model. Pandas and NumPy handle data preprocessing. Joblib saves and loads the trained model. Matplotlib and Seaborn generate data visualizations and analysis charts.